AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 24 · WORKBOOK

Reproducible environment and records

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Link exact inputs, script, runtime, metric and raw output without confusing replay with scientific validity.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

Power and resource tradeoffs

> **My remembered idea, example and correction**

Predefining an analysis plan

> **My remembered idea, example and correction**

Dataset and benchmark limitations

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A reproducible run record links exact data IDs/splits and bytes, code and preprocessing versions, dependency/runtime versions, settings, randomness policy, metric and raw outputs. Seed 42 plus final-score.txt alone cannot reconstruct the main scenario because those materials are absent. A seed cannot generate missing code or data and may not control every random process.

  Separate supplied mean example uses 2, 4, 6: sum twelve divided by three gives four. Changed values 2, 4, 9 give fifteen divided by three, or five. Retain original and changed inputs with separate run IDs. The self-contained code actually executes only this small in-memory calculation, hashes its exact UTF-8 input/script bytes and reports its actual Python runtime. No randomness is used.

  Byte integrity is not a proof of source provenance, valid labels, sampling or the right scientific method. New preprocessing or data versions require preserved original records and a new linked run record. No independent AI experiment is reproduced by this demonstration.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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- Folder contains seed 42 and final-score.txt, but no dataset, code or dependency record. Show the intermediate steps and give the conclusion with its scope.

Main supplied artifact	What it tells us	What remains missing
Seed 42	A stated seed value	Algorithm, input, random library and how seed was used
final-score.txt	A final output record exists in the scenario	Which metric, denominator and run generated it
No dataset/split	Cannot identify evaluated cases	Exact data materials and roles
No code/preprocessing version	Cannot reconstruct transformation	Exact executable procedure

Main supplied artifact	What it tells us	What remains missing
No dependencies/runtime	Cannot establish executable environment	Versions and execution instructions
Replay claim	Not established	Supply artifacts and perform a documented check

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

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- A report omits this qualification: Reproducibility supports checking the computation, not automatic validity of the research claim.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A reproducible run record links exact data IDs/splits and bytes, code and preprocessing versions, dependency/runtime versions, settings, randomness policy, metric and raw outputs. Seed 42 plus final-score.txt alone cannot reconstruct the main scenario because those materials are absent. A seed cannot generate missing code or data and may not control every random process.

  Separate supplied mean example uses 2, 4, 6: sum twelve divided by three gives four. Changed values 2, 4, 9 give fifteen divided by three, or five. Retain original and changed inputs with separate run IDs. The self-contained code actually executes only this small in-memory calculation, hashes its exact UTF-8 input/script bytes and reports its actual Python runtime. No randomness is used.

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- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Partner changes pre**process**ing and data version while keeping old run ID. What must be preserved?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled reproducible environment and records **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of reproducible environment and records.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Reproducible environment and records. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Reproducible environment and records. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Partner changes pre**process**ing and data version while keeping old run ID. What must be preserved?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

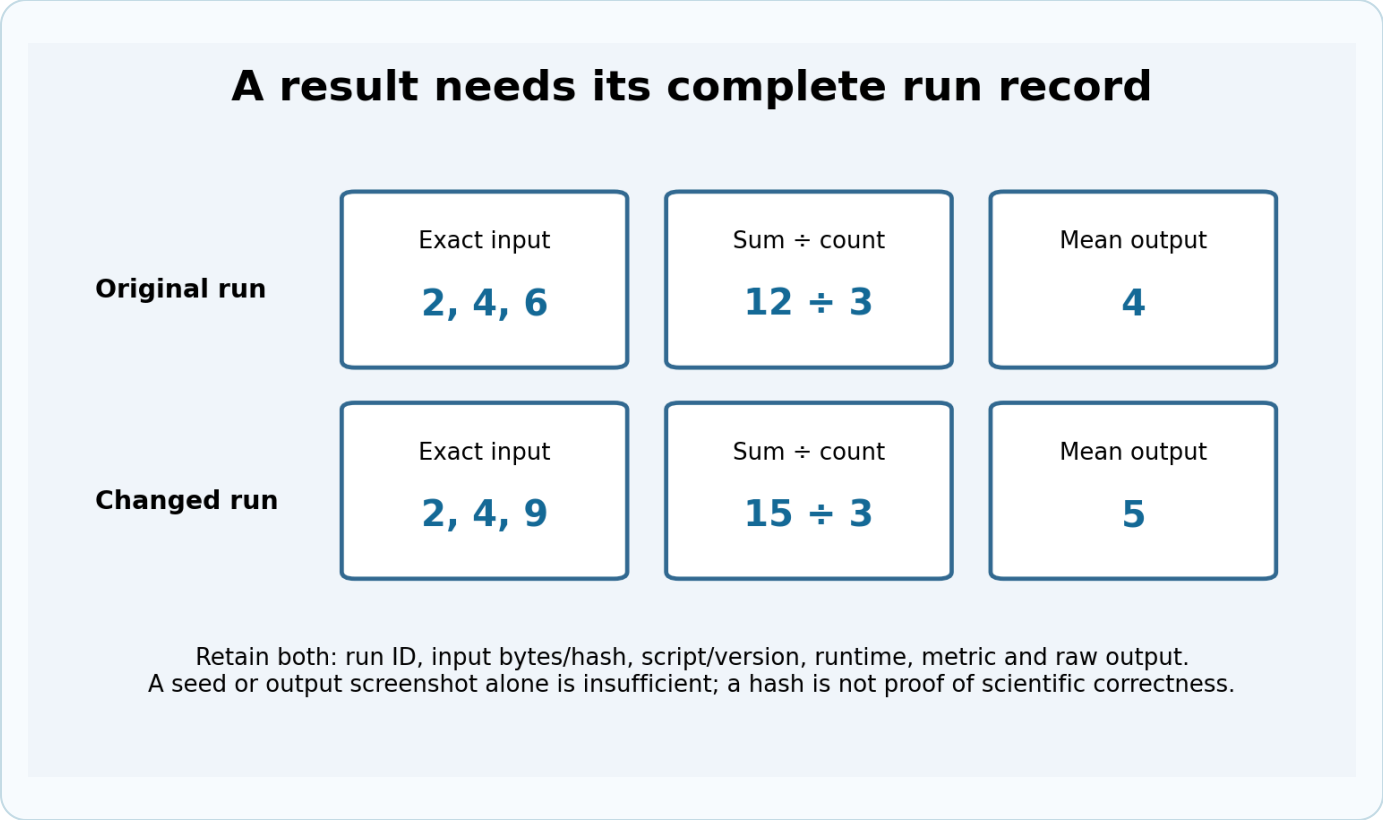
> Keep this task's inputs,rule and response together.

Complete task and evidence

- A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version.
- Data file contains 2,4,6. Recorded script sums values and divides by count; environment and file hash are saved.
- Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the **output** so another team can identify the exact run?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/24

Recorded mean calculation

	Value 1	Value 2	Value 3	Mean
Original data	2	4	6	4
Changed data	2	4	9	5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/24

Reproducible environment and records

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify exact input

Use the three recorded values, not a similarly named edited file.





2. Trace calculation

Sum is 12; count is 3; mean is 4.





3. Save provenance

Retain script version, environment description and data hash with the output.

CHANGE ONE THING

The data file changes to 2,4,9.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version.
- Which labelled step matches this detail? Retain script version, environment description and data hash with the **output**.
- 1. Trace calculation
- 2. Identify exact **input**
- 3. Save provenance
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Save provenance Retain script version, environment description and data hash with the **output**.

Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

Changed-case question and optional hint

- Changed-case practice: The data file changes to 2,4,9. Which conclusion is supported by the supplied facts?
- 1. The mean becomes 15/3=5, and the saved **input** hash must change.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The mean becomes 15/3=5, and the saved **input** hash must change. The calculation is the same but its **input** records differ.

Repair and retain the record

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the **mechanism** and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the **mechanism** and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

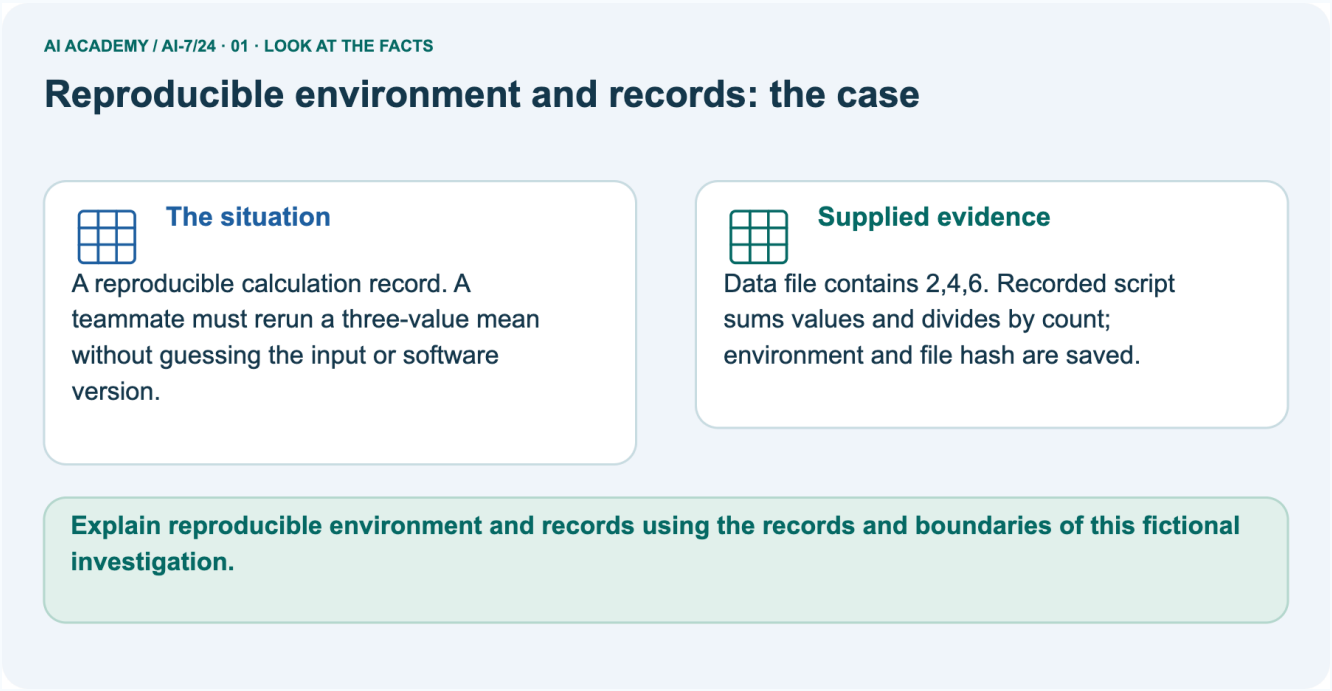
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Reproducible environment and records: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

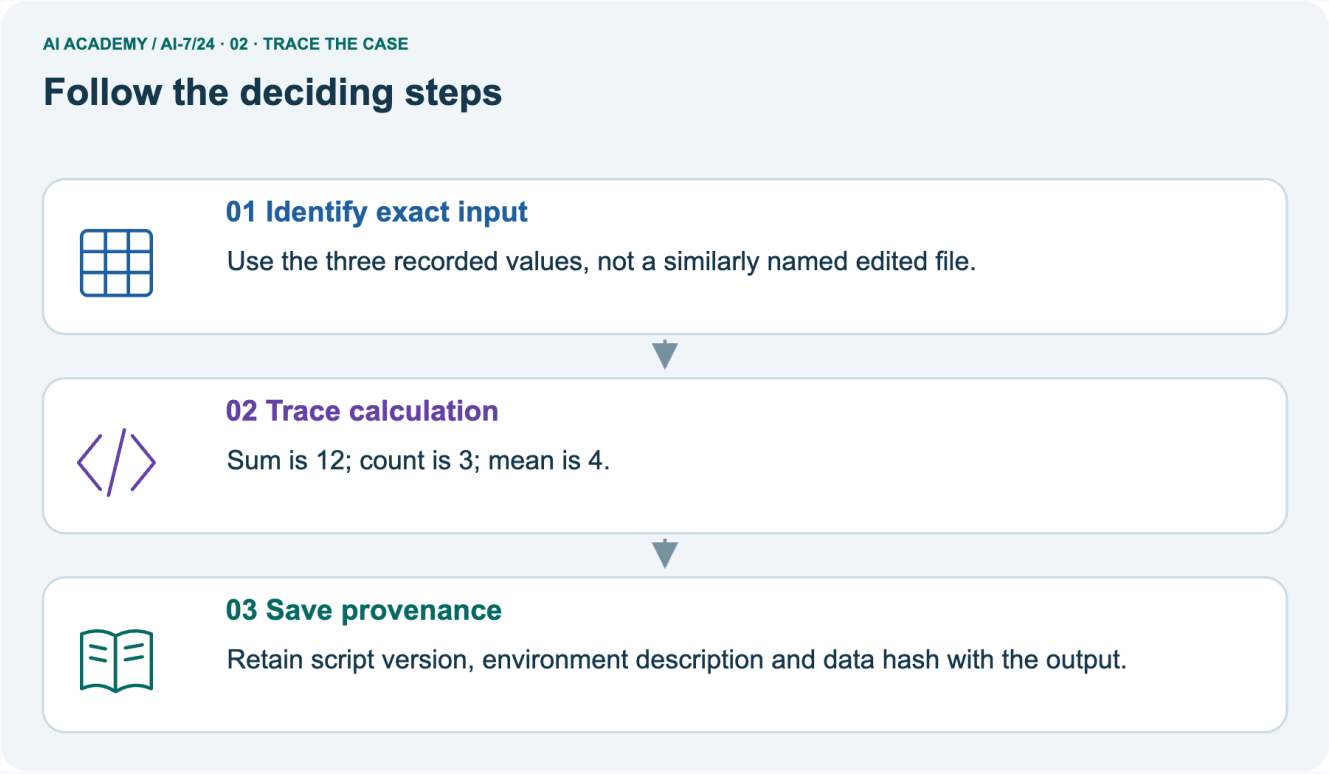
A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version. Supplied facts: Data file contains 2,4,6.

Recorded script sums values and divides by count; environment and file hash are saved.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



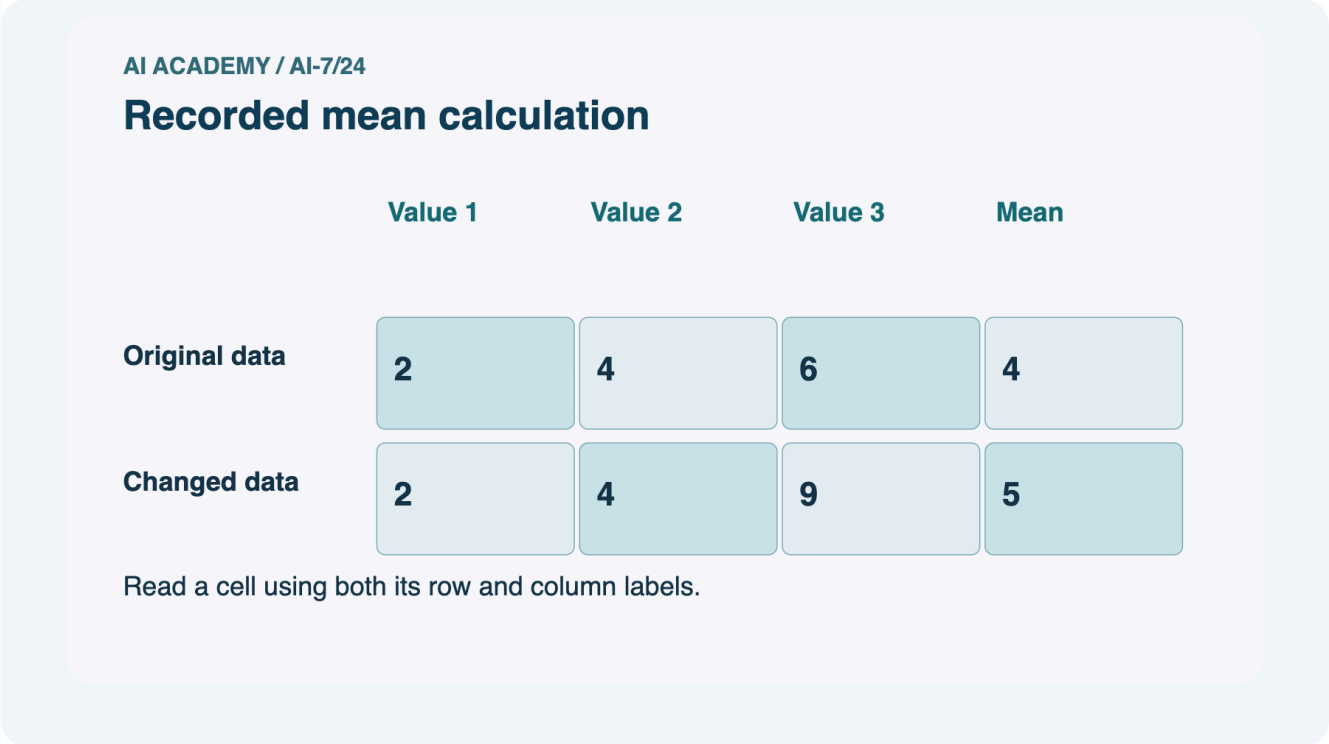
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify exact **input**: Use the three recorded values, not a similarly named edited file. → Trace calculation: Sum is 12; count is 3; mean is 4. → Save provenance: Retain script version, environment description and data hash with the **output**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Recorded mean calculation



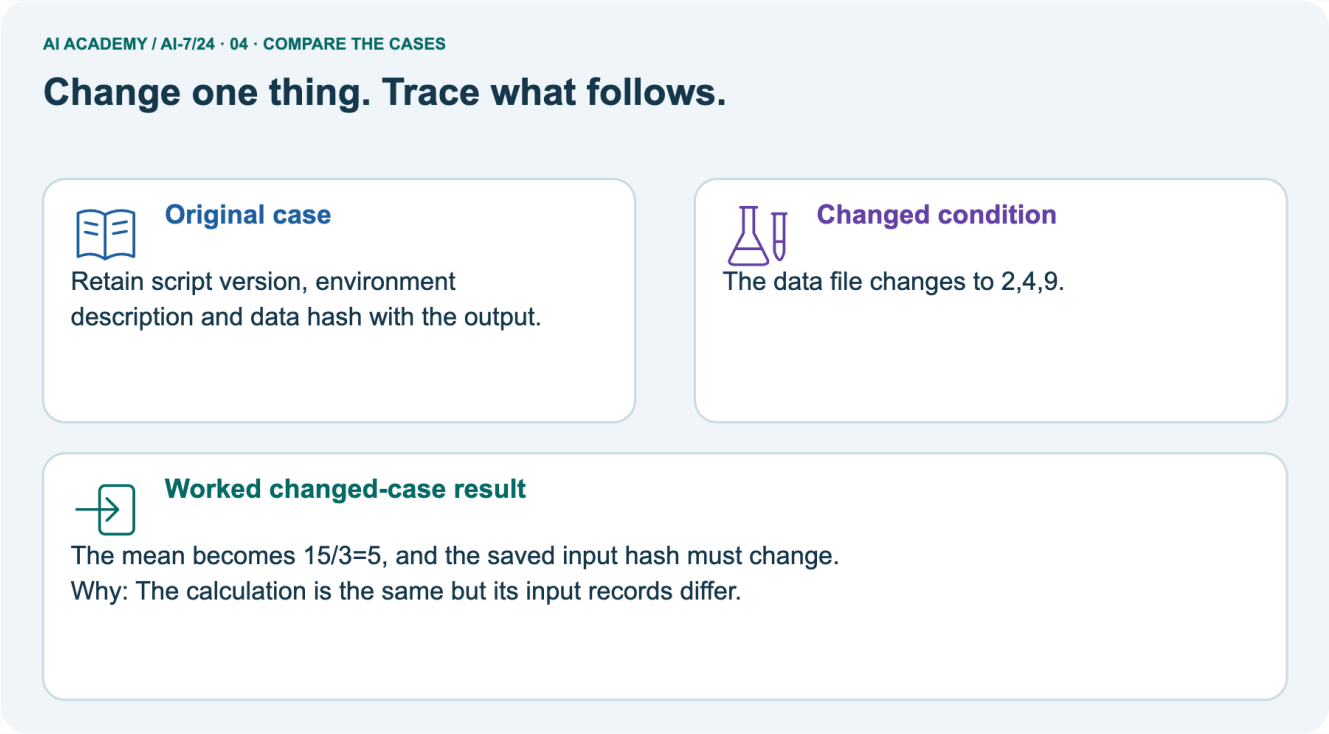
Follow the labelled picture with the matching explanation. Open the vector diagram

Recorded mean calculation. Original data: Value 1 = 2, Value 2 = 4, Value 3 = 6, Mean = 4. Changed data: Value 1 = 2, Value 2 = 4, Value 3 = 9, Mean = 5.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

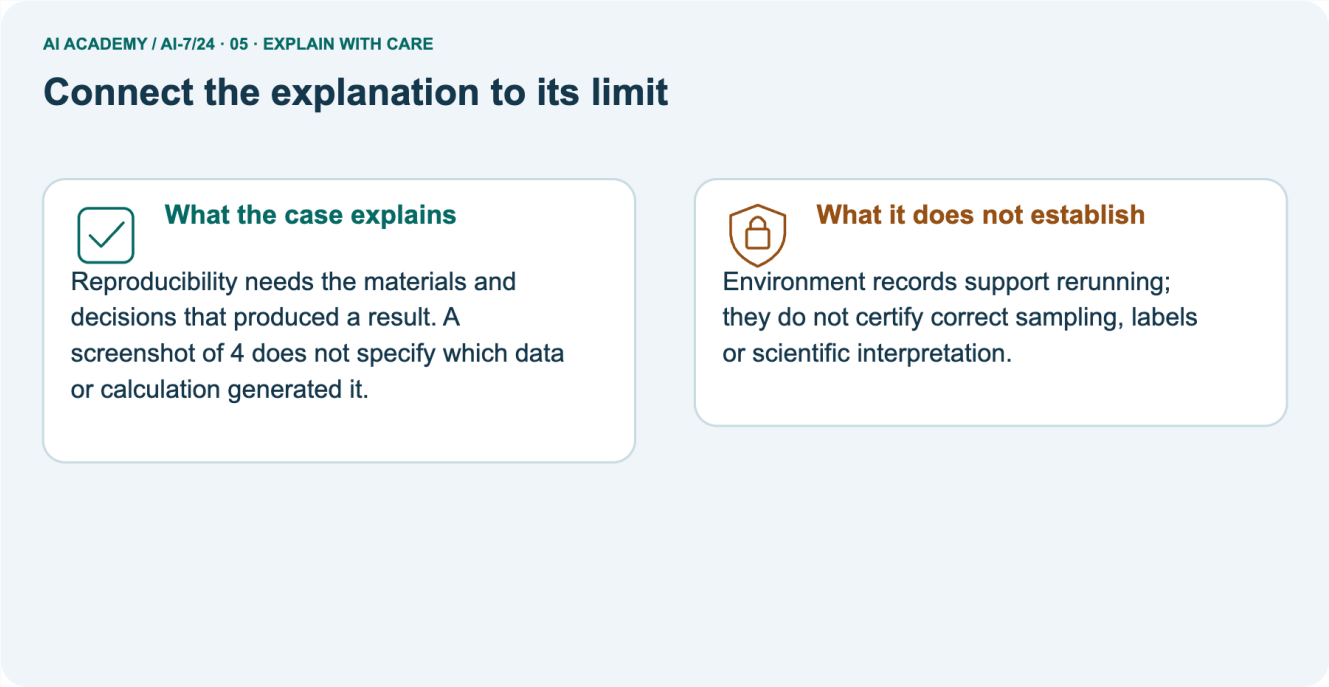
Original case: Retain script version, environment description and data hash with the **output**. Changed condition: The data file changes to 2,4,9. Expected result: The mean becomes 15/3=5, and the saved **input** hash must change.

Reason: The calculation is the same but its **input** records differ.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

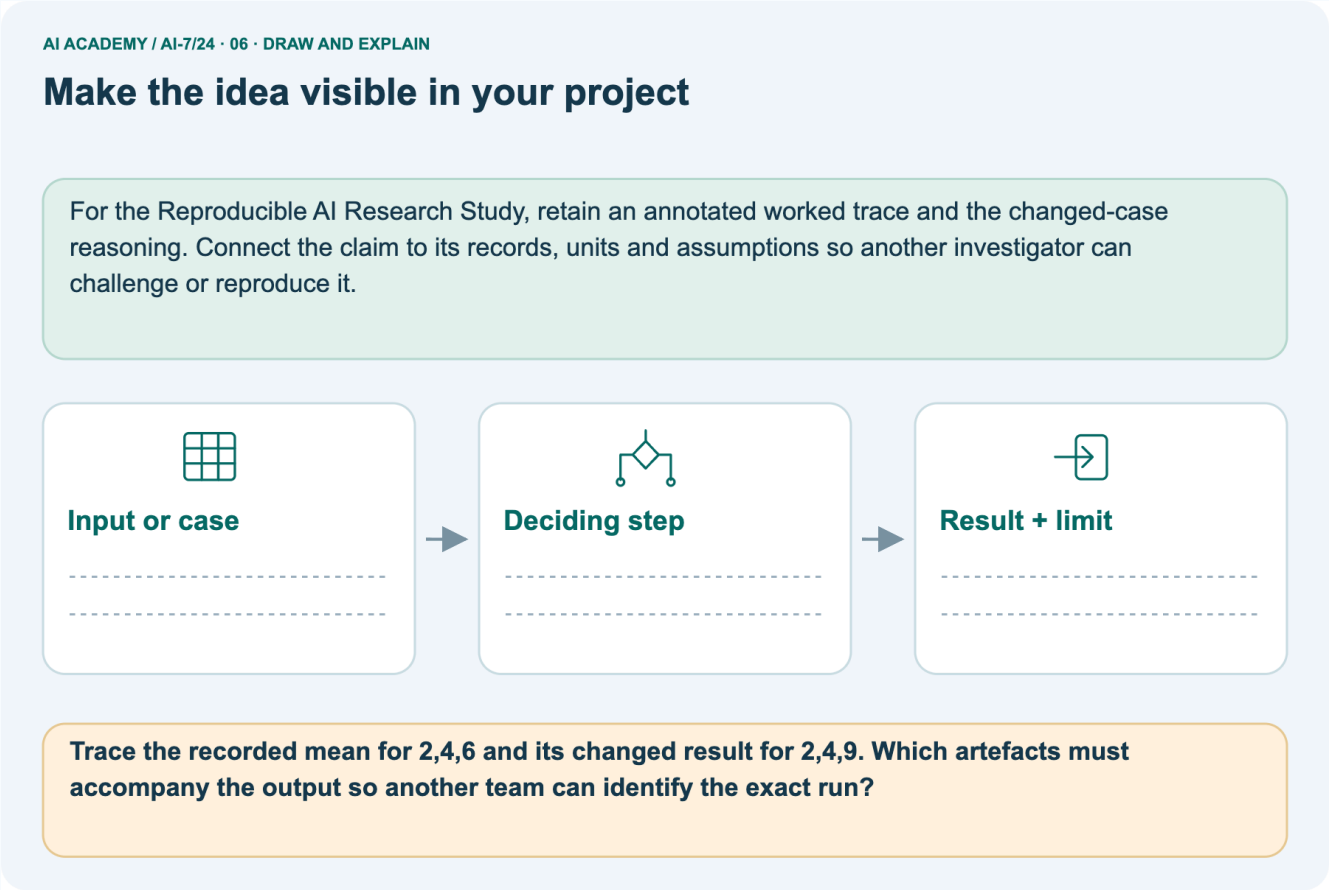
Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

Limit: Environment records support rerunning; they do not certify correct sampling, labels or scientific interpretation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the **output** so another team can identify the exact run?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Reproducible environment and records: the case

02. Follow the deciding steps

03. Recorded mean calculation

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.